

Tensors

A tensor is just a multi-dimensional array of numbers. It is a container that can hold data in any number of dimension.

Tensors are categorized by their Rank

Rank	Name	Visual Analog	Description
0	Scalar	A single point	Single no. eg. [7]
1	Vector	A line	A list of no. [7, 1.2, 5]
2	Matrix	A grid	A table with rows & columns
3+	Tensor	A cube (or higher)	A stack of matrices.

Example of 1D Tensors

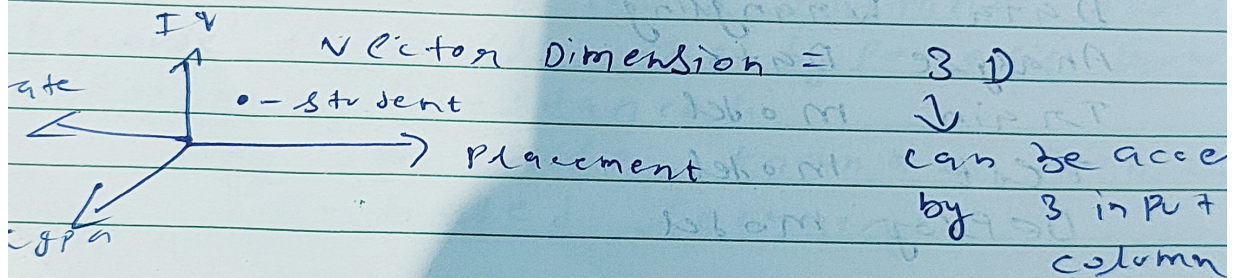
Students 10000

cgpa | i | state | Placement

8.1 91 K.B 1

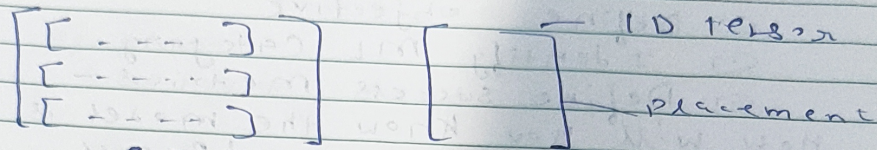
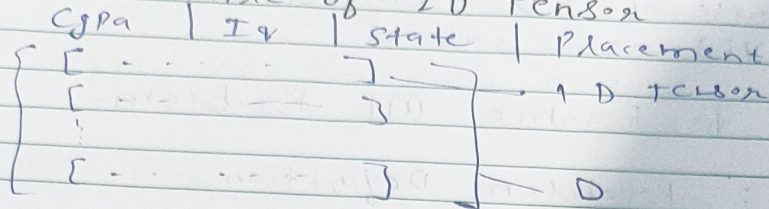
(0)

1D tensor = [8.1, 91, 0]





Example of 2D Tensor



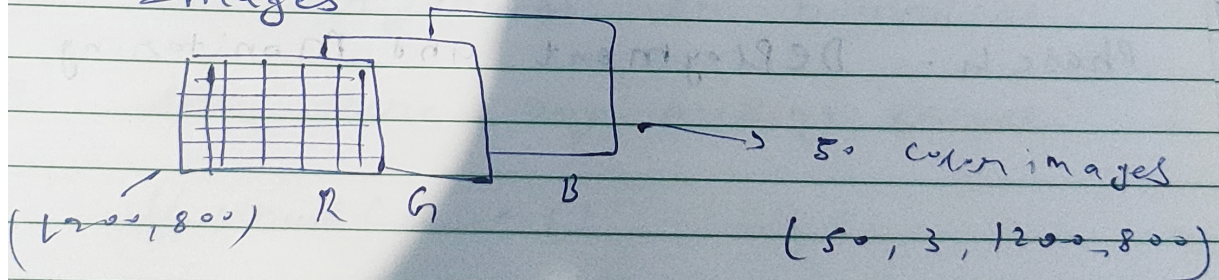
Example of 3D Tensor

Hi	Shubham	Hi	Nitish	Rahul	R
Hi	Rahul	Hi	Shubham	Rahul	Agastya
Hi	Agastya	0	1	0	0
[[1,0,0,0],[0,1,0,0]]		0	0	1	0
[[1,0,0,0],[0,0,1,0]]		0	0	0	1
[[1,0,0,0],[0,0,0,1]]		3D tensor			

Most of them are in time series
In Medical Domain, Stock Market Data

Example of 4D Tensor

Images





Example of SD tensors
Videos - frames

How to Frame ML Problem?

Phase 1: Problem Definition

- ① Define objective
- ② Identify ML Category
- ③ Define success metrics -

How will you know the model is good? Use Precision / Recall, mean Absolute error or click-through rate

Phase 2: Data Strategy

- ① Data Sourcing
- ② Selection of Features
- ③ Labelling
- ④ Ethical check Are biases in data

Phase 3: Project Roadmap

- ① EDA
- ② Feature Engineering
- ③ Baseline model
- ④ Evaluation and iteration

Phase 4: Deployment and Monitoring



Understanding Your Data

Q What are the basic question you ask when you get a dataset.

① How big is the Data?

`df.shape`

It will give rows and column.

② How does the data look like?

`df.head()`

It will give top five rows of data just to give overview of it.

`df.sample(5)` will give 5 random rows from dataset.

③ What is the Datatype of cols?

`df.info()`

It will give basic information of each column, memory space, null/not null, Datatype, Column is categorical or numerical.

④ Are there any missing values?

`df.isnull().sum()`

⑤ How does Data look mathematically?

`df.describe()`



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⑥ Are there any duplicate values?

`df.duplicated().sum()`

⑦ How is the correlation b/w cols?

Which input column impact the output column, If there is no input column make an impact to output we simply remove it.

`df.corr()`

Univariate Analysis

Suppose you have multiple columns independent Analysis